

such as 3G and LTE. Cellular standards continue to evolve and, although LTE has been around for a while now, design engineers now need to start looking at LTE Advanced as well," adds Hire.

Ability to test dynamic equipment.

The functionality of signal generators was getting defined in software which was good enough for performing test on electronic circuits of the 20th century. "This evolution required a different set of functions to be generated at very different speeds for different kinds of models for different environments," informs Satish Mohanram, technical marketing manager, National Instruments India.

For environments where equipment functionality is very dynamic, ensuring that the functionality of these equipment gets tested becomes a lot more complicated. One has to basically make the test system behave as if it is the real-world environment for that specific application.

Citing an example, Mohanram explains, "If you consider ECUs, you have to make the environment work as though it is the actual part, be it the transmission, the engine or the terrain on which it is going. This really brings in a lot of complexity into the type of signals that have to be generated; signals have to be generated really fast and change really quick. Software was the key but it was not providing the required speed and functionality wherever it was needed. This moved us into the next generation which is a software-defined signal generator having an on-board signal processing unit."

Programmable signal generators.

"These signal generators are where we stand currently and I think, with the kind of advancements that are happening in the field-programmable gate arrays (FPGAs), there is a lot of functionality that gets added to the signal generators in a way that they can generate or simulate different scenarios," believes Mohanram.

In the past, FPGA was a way of incorporating your own intellec-

Which new technologies in signal generators are helping design engineers?

1. Digital up-conversion:
 - (a) Helps generate signals in desired frequency bands.
 - (b) Avoids some problems associated with analogue I/Q modulators:
 - (i) No (in-band) carrier feed-through
 - (ii) No (in-band) images
2. Direct digital frequency synthesis: Helps users create very fast switching signals for applications such as frequency hopping, pulse generation, etc.
3. Support for higher modulation bandwidths: In order to provide higher throughput, longer range and better resolution, modulation bandwidth requirements are constantly increasing. Modulation bandwidths up to 2GHz are available today; however, future developments will see even higher bandwidth requirements.
4. Creation of custom signal formats: Many evolving communication standards require specific signal types to be generated. Also, defence requirements work on their specific and non-conventional signal types, which means they need to generate signals that are non-custom signals.
5. Recreation of a captured off-the-air signal: Defence applications require their receivers to be tested under real-time signals, which means they need to create real scenario to test their receiver. For example, if a RADAR receiver needs to be tested, one needs to fly aircraft to validate the receiver performance.

Data courtesy: Vishal Gupta, senior application consultant (RF/MW, Surveillance), Agilent Technologies

tual property (IP) into any system as it is programmable. With this, you get great performance and at the same time have your functionality implemented. He adds, "FPGAs moved into the signal generators around a year or a year-and-a-half ago, which I believe is a significant move that has occurred whereby the functionality of signal generators has grown by leaps and bounds."

New technologies that benefit the engineer. The functionality that gets into today's designs is very much defined in software, which is a challenge faced by most of the design engineers today. As software evolves, being able to set different types of scenarios becomes very difficult. Earlier it was very simple because you could give a square or a sine wave at the input and see how the response was on the other side. But today the algorithms that are implemented over these devices also change in real time.

Citing an example, Mohanram says, "Consider an adaptive noise cancellation system. In such a system, when you give a certain input, the output will not be the same but it will vary based on the ambient. For you to check that, the ambient needs to be simulated first and then the actual

input signal. The actual signal is simple but the ambient is pretty dynamic in nature. One has to basically frame the mathematical model of ambient and get that to run."

These kind of things were never possible but, with the arrival of programmable signal generators, these have become possible because the model can be hard coded into the hardware. And it gets implemented in real time, unlike the earlier generation where if implemented in software would result in lags and inconsistencies. Hence, it provides design engineers the capability to comprehensively test the functionality, say, build better quality products for the market.

"Also, today's signal generators have two outputs on a single instrument," informs Madhukar Tripathi, regional manager, Anritsu India Pvt Ltd. He says, "This is very good for many applications where usage of two instruments and syncing them is necessary. Two-channel concept has eliminated the need for a second instrument, hence cost has receded, which leads to more accurate and faster measurements."

Design engineers can now use these signal generators and make faster and accurate measurements. Local